NexERP — Panduan Pengguna Modul Manufacturing

- **Versi dokumen:** 1.0 · **Tanggal:** 6 November 2026
- **Sistem:** NexERP (berbasis ERPNext / Frappe) — `https://nexerp.nexoratech.co/desk/manufacturing`
- **Sumber analisa:** kode repository `imam-nurokhi/erpnext` (branch rebrand NexERP) + tampilan halaman Manufacturing di situs live.
- **Ruang lingkup:** hanya modul **Manufacturing** (termasuk keterkaitan langsung ke Stock, Subcontracting, dan Quality).

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1. Pendahuluan

Modul Manufacturing di NexERP dipakai untuk mengelola seluruh siklus produksi:

- **Definisi produk**: struktur bahan (BOM), operasi, mesin/workstation, routing.
- **Perencanaan**: peramalan permintaan, Master Production Schedule (MPS), Material Requirements Planning (MRP), Production Plan.
- **Eksekusi**: Work Order, Job Card (pencatatan waktu kerja per operasi), mutasi stok produksi.
- **Pengendalian**: downtime mesin, quality inspection, process loss, biaya produksi, laporan.

Semua transaksi produksi otomatis terhubung ke **Stock** (mutasi barang) dan **Accounting** (nilai persediaan & biaya), sehingga Harga Pokok Produksi terbentuk tanpa input manual tambahan.

1.1 Istilah Penting

Istilah	Arti
Raw Material (RM)	Bahan baku yang dikonsumsi dalam produksi
Sub-assembly / Semi-Finished Good	Barang setengah jadi yang punya BOM sendiri
Finished Good (FG)	Barang jadi hasil produksi
BOM	Bill of Materials — resep/daftar bahan & operasi untuk membuat 1 batch barang
Work Order (WO)	Perintah produksi untuk sejumlah qty barang tertentu
Job Card	Kartu kerja per operasi per WO; tempat mencatat waktu & qty selesai
Workstation	Mesin / lini / meja kerja tempat operasi dilakukan
WIP Warehouse	Gudang Work-in-Progress (bahan yang sedang diproses)
Backflush	Pemotongan stok bahan baku otomatis saat barang jadi dicatat
Process Loss	Susut proses (qty yang hilang selama produksi)
Secondary Items	Scrap / co-product / by-product yang muncul dari produksi
MPS / MRP	Master Production Schedule / Material Requirements Planning

2. Peran Pengguna & Hak Akses

Role	Hak utama
Manufacturing Manager	Membuat & submit BOM, mengubah Manufacturing Settings, approve perencanaan
Manufacturing User	Membuat Work Order, Production Plan, Job Card, BOM
Stock User	Membuat Work Order & Stock Entry (transfer/manufacture)
Operator	Akses halaman Visual Plant Floor (start/complete job)
System Manager	Semua, termasuk Job Card

> **Rekomendasi:** pisahkan role *Planner* (Production Plan/MPS), *Supervisor* (Work Order), *Operator* (Job Card saja) dan *Gudang* (Stock Entry). Lihat Gap G-11.

3. Mengenal Halaman Manufacturing

URL: `/desk/manufacturing`

3.1 Sidebar Kiri

Menu	Isi
Home	Halaman workspace (chart, kartu angka, shortcut)
Dashboard	Dashboard Manufacturing (chart analitik lebih lengkap)
BOM	Daftar Bill of Materials

Menu	Isi
Work Order	Daftar perintah produksi
Job Card	Daftar kartu kerja operasi
Stock Entry	Mutasi stok (transfer bahan, manufacture)
Material Planning ▸	Item Lead Time, Production Plan, Forecasting, Master Production Schedule, Sales Forecast, Production Planning Report
Tools ▸	BOM Creator, BOM Update Tool, BOM Comparison Tool, Downtime Entry
Reports ▸	10 laporan produksi (lihat Bab 10)
Setup ▸	Item, Warehouse, Operation, Workstation, Workstation Type, Plant Floor, Routing
Settings	Manufacturing Settings
Getting Started	Panel onboarding 7 langkah

3.2 Area Utama (Home)

Widget	Sumber data	Arti
Produced Quantity (chart garis)	Work Order submit, field `produced_qty`	Total qty diproduksi per periode (filter: Last Year, Monthly)
Open Work Orders	Work Order status *Not Started*	WO yang sudah disubmit tapi belum mulai
WIP Work Orders	Work Order status *In Process*	WO yang sedang berjalan
Manufactured Items Value	Stock Entry purpose *Manufacture*, bulan ini	Nilai (Rp) barang jadi yang masuk gudang bulan ini
Kartu Link	Bill of Materials, Production, Subcontracting, Reports, Tools, Settings	Akses cepat ke dokumen terkait

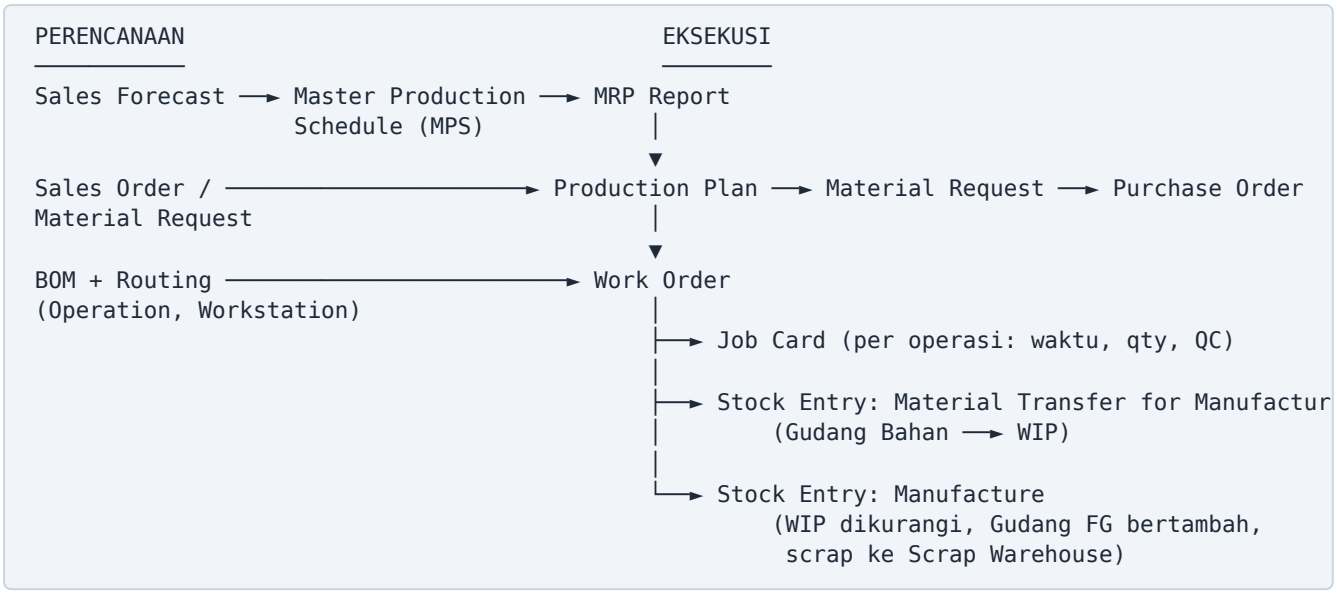
3.3 Getting Started (Onboarding)

#	Langkah	Aksi
1	Create Raw Material	Buat Item bahan baku
2	Create Finished Good	Buat Item barang jadi
3	Create Operation	Buat master Operation
4	Create Bill of Materials	Buat BOM
5	Create Work Order	Buat Work Order
6	View Work Order Summary	Buka laporan Work Order Summary
7	Review Manufacturing Settings	Cek & simpan Manufacturing Settings

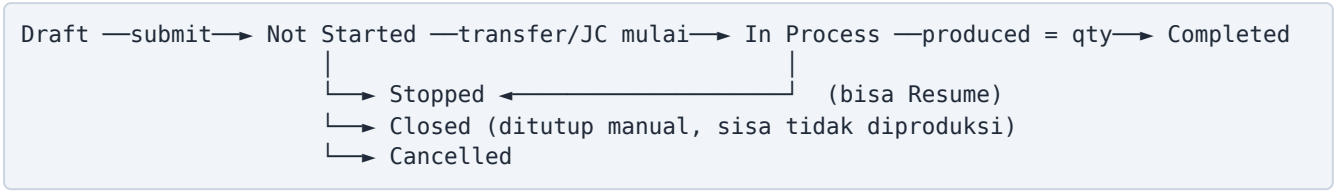
Progres saat ini di situs live: **0/7 (0%)**.

4. Konsep & Alur Proses Produksi

4.1 Alur Besar



4.2 Siklus Status Work Order



Status tambahan: **Stock Reserved / Stock Partially Reserved** (jika reservasi stok aktif).

4.3 Siklus Status Job Card

`Open → Work In Progress → (On Hold) → Material Transferred → Completed`, lalu Submitted / Cancelled.

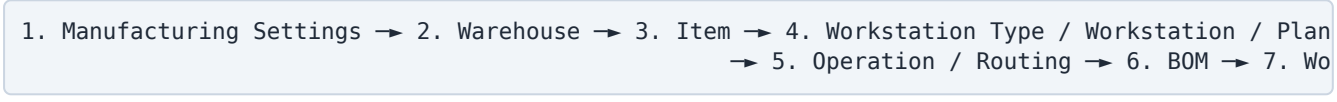
4.4 Dampak Akuntansi (ringkas)

Transaksi	Debit	Kredit
Transfer bahan ke WIP	Persediaan WIP	Persediaan Bahan Baku
Manufacture	Persediaan Barang Jadi	Persediaan WIP / Bahan Baku + Biaya Operasi (dibebankan)

Biaya operasi berasal dari *hour rate* Workstation × waktu operasi (BOM atau Job Card aktual).

5. Setup Awal (Langkah demi Langkah)

Urutan yang **wajib** diikuti sebelum transaksi pertama:



5.1 Manufacturing Settings

Menu: **Settings** (sidebar paling bawah). Hanya Manufacturing Manager/System Manager.

Tab *BOM and Production*

Pengaturan	Default	Rekomendasi awal	Penjelasan
Backflush Raw Materials Based On	BOM	**BOM**	*BOM*: bahan dipotong sesuai resep. *Material Transferred for Manufacture*: dipotong sesuai yang benar-benar ditransfer ke WIP
Allow Continuous Material Consumption	Off	Off	Jika On, bahan bisa dikonsumsi bertahap sebelum barang jadi dicatat
Update BOM Cost Automatically	Off	**On**	Scheduler memperbarui biaya BOM dari harga terbaru
Make Serial No / Batch from Work Order	Off	Sesuai kebutuhan	Serial/batch barang jadi dibuat otomatis saat WO disubmit
Add Corrective Operation Cost in FG Valuation	Off	On (jika ada rework)	Biaya rework ikut menambah nilai barang jadi
Validate Components and Quantities Per BOM	Off	**On** (disiplin)	Stock Entry ditolak jika tidak sesuai BOM
Transfer Extra Raw Materials to WIP (%)	0	5–10%	Toleransi transfer bahan lebih
Allow Editing of Items and Quantities in Work Order	Off	Off	Izinkan ubah bahan di WO
Set Operating Cost / Secondary Items From Sub-assemblies	Off	Sesuai kebutuhan	Biaya operasi & scrap sub-assembly ikut ke FG tanpa Job Card
Overproduction % for Sales Order / Work Order	0	0–5%	Toleransi produksi melebihi pesanan

Tab *Job Card and Capacity Planning*

Pengaturan	Default	Penjelasan
Disable Capacity Planning	Off	Matikan jika tidak ingin penjadwalan otomatis per workstation
Capacity Planning For (Days)	30	Horizon penjadwalan operasi
Time Between Operations (Mins)	10	Jeda antar operasi
Allow Overtime	Off	Boleh menjadwalkan di luar jam kerja workstation
Allow Production on Holidays	Off	Boleh menjadwalkan di hari libur (Holiday List)
Allow Excess Material Transfer	Off	Transfer bahan di Job Card setelah kebutuhan terpenuhi
Get Raw Materials Cost from Consumption Entry	Off	Biaya RM dari entri konsumsi
Enforce Time Logs	Off	Time log Job Card wajib From/To Time

Default Warehouse (WIP, FG, Scrap) juga diatur di sini dan dipakai otomatis oleh Work Order baru.

5.2 Warehouse

Buat minimal gudang berikut (per Company):

Gudang	Fungsi
Stores / Gudang Bahan Baku	Sumber bahan baku
Work In Progress	Bahan yang sedang diproses
Finished Goods	Hasil produksi
Scrap (opsional)	Sisa produksi / scrap

5.3 Item

Jenis	Setting penting
Bahan Baku	`Maintain Stock` = Ya, Item Group (mis. *Raw Material*), UOM, Valuation Rate awal, Default Warehouse
Sub-assembly	`Maintain Stock` = Ya, nanti diberi BOM
Barang Jadi	`Maintain Stock` = Ya, Default BOM (otomatis terisi saat BOM default disubmit)
Serial / Batch	Centang *Has Serial No* / *Has Batch No* jika perlu ketelusuran
Subkontrak	Centang *Is Sub Contracted Item* untuk bahan yang dikerjakan vendor

5.4 Workstation Type, Workstation, Plant Floor

1. **Workstation Type** — kelompok mesin (mis. *Mesin CNC*, *Meja Assembly*). Isi *Hour Rate* & komponen biaya (listrik, sewa, operator).
2. **Workstation** — mesin spesifik: - *Workstation Type*, *Plant Floor*, *Warehouse* - *Production Capacity* (jumlah job paralel) - *Working Hours* (jam kerja per hari) + *Holiday List* - *Workstation Costs* → menghasilkan **Hour Rate** (Rp/jam) - *Status*: Production / Off / Idle / Problem / Maintenance / Setup
3. **Plant Floor** — denah lantai produksi. Di form Plant Floor tampil **Visual Plant Floor**: status semua workstation, Job Card aktif, stok di gudang lantai, serta tombol Start/Complete job dan buat Stock Entry.

5.5 Operation & Routing

- **Operation**: nama proses (Potong, Rakit, Cat, QC).
- *Default Workstation*, *Sub Operations* (langkah detail + waktu)
- *Batch Size* + *Create Job Card based on Batch Size*
- *Is Corrective Operation* (untuk rework)
- *Quality Inspection Template*
- **Routing**: urutan operasi standar (Operation + Workstation + waktu menit + batch size). Satu Routing bisa dipakai banyak BOM; biaya operasi dihitung otomatis.

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6. Bill of Materials (BOM)

6.1 Membuat BOM (cara standar)

1. Menu **BOM → + Add BOM**.
2. Isi **Item** (barang jadi/sub-assembly) dan **Quantity** (qty hasil per BOM, mis. 1 atau 100).
3. Bagian **Raw Materials**: tambah item bahan, qty, UOM. Rate terisi otomatis sesuai *Rate Of Materials Based On*: - *Valuation Rate*, *Price List*, *Last Purchase Rate*, atau *Manual*.
4. (Opsional) Centang **With Operations** → pilih **Routing** atau isi tabel **Operations** (Operation, Workstation, Time in mins).
5. (Opsional) **Secondary Items** — scrap/co-product beserta % alokasi biaya (total alokasi FG + secondary harus 100%).
6. (Opsional) **Process Loss %**, **Inspection Required**, **Allow Alternative Item**, **Project**.
7. **Save → Submit**. Centang **Is Active** dan **Is Default** untuk BOM utama.

Aturan sistem: - Hanya **satu BOM default** per item; BOM default otomatis tercatat di Item. - BOM yang dipakai BOM lain **tidak bisa dinonaktifkan/dicancel** (gunakan BOM Update Tool → Replace). - Mengubah BOM yang sudah submit = **Amend/Copy** menjadi versi baru (penomoran `BOM-ITEM-001`, `-002`, ...). - Biaya total = Raw Material Cost + Operating Cost (mendukung multi currency).

6.2 Multi-level BOM

Jika bahan adalah sub-assembly yang punya BOM, field **BOM No** di baris bahan terisi otomatis. Tabel **Exploded Items** menampilkan kebutuhan bahan baku sampai level terbawah. Di Work Order, centang **Use Multi-Level BOM** untuk langsung memakai bahan level terbawah (tanpa WO sub-assembly).

Phantom BOM: BOM yang tidak diproduksi tersendiri — bahannya langsung "diledakkan" ke BOM induk.

6.3 BOM Creator (membuat BOM bertingkat sekaligus)

Menu **Tools → BOM Creator**: 1. Isi Finished Good, Qty, Company, *Rate Based On*. 2. Pada tampilan pohon, tambah bahan; klik bahan → **Add Sub Assembly** untuk membuat level di bawahnya. 3. **Submit** → sistem membuat semua BOM (dari level terbawah) di background. Status: *In Progress* → *Completed* (atau *Failed* dengan Error Log).

6.4 BOM Update Tool

Menu **Tools → BOM Update Tool**: - **Replace BOM**: pilih *Current BOM* dan *New BOM* → ganti di semua BOM induk. Dijalankan sebagai background job. - **Update latest price in all BOMs**: hitung ulang biaya seluruh BOM level demi level. - Riwayat & status job tercatat di **BOM Update Log** (Queued, In Progress, Completed, Failed).

6.5 BOM Comparison Tool

Menu **Tools → BOM Comparison Tool**: pilih dua BOM → tampil perbedaan item, qty, rate, dan operasi. Berguna untuk audit perubahan resep.

7. Eksekusi Produksi

7.1 Work Order — Langkah demi Langkah

1. **Work Order → + Add** (atau dari Sales Order / Production Plan / Material Request).
2. Isi: - **Item To Manufacture**, **BOM No** (default terisi otomatis), **Qty To Manufacture** - **Source / WIP / Target (FG) / Scrap Warehouse** - **Planned Start Date**, **Expected Delivery Date** - Opsi: *Use Multi-Level BOM*, *Skip Material Transfer to WIP*, *Reserve Stock*, *Project*, *Sales Order*
3. Tabel **Required Items** (bahan) dan **Operations** terisi dari BOM.
4. **Submit** → status **Not Started**. Jika ada operasi & capacity planning aktif, jadwal operasi dihitung dan **Job Card** dapat dibuat.
5. **Start** → membuat Stock Entry *Material Transfer for Manufacture* (Stores → WIP). *(Dilewati bila "Skip Material Transfer" aktif.)*
6. Kerjakan operasi melalui **Job Card** (7.2).
7. **Finish** → membuat Stock Entry *Manufacture*: bahan dikonsumsi (backflush), barang jadi masuk FG Warehouse, scrap masuk Scrap Warehouse.
8. Saat Produced Qty = Qty, status **Completed**.

Tombol/aksi lain di Work Order:

Aksi	Fungsi
Create Job Card	Membuat Job Card per operasi
Create Pick List	Daftar ambil bahan untuk gudang
Stop / Re-open	Menghentikan sementara WO
Close	Menutup WO walaupun belum selesai (sisa tidak diproduksi)
Disassemble	Membongkar barang jadi kembali menjadi komponen
Return Components	Mengembalikan sisa bahan dari WIP ke gudang
Reserve / Unreserve Stock	Reservasi stok bahan untuk WO
Create BOM	Membuat BOM dari WO (jika belum ada)

7.2 Job Card

Dibuat per **operasi** dari Work Order.

1. Buka Job Card → **Start Job** (pilih Employee). Timer berjalan.
2. **Pause / Resume** jika ada jeda (status *On Hold*).
3. **Complete Job** → isi *Completed Qty* (dan *Process Loss Qty* jika ada).
4. Jika operasi butuh bahan khusus, gunakan **Material Request** atau **Material Transfer** dari Job Card.
5. Jika *Inspection Required*, buat **Quality Inspection** sebelum submit.
6. **Submit** → completed qty & waktu aktual diperbarui ke Work Order; biaya operasi aktual = hour rate × waktu aktual.

Fitur lanjutan: - **Sub Operations** — checklist langkah dalam satu operasi. - **Corrective Job Card** — untuk rework (operasi berflag *Is Corrective Operation*). - **Subcontracting PO per operasi** — operasi tertentu dikerjakan vendor. - **Semi-Finished Goods tracking** — hasil per operasi dicatat sebagai barang setengah jadi. - Validasi **overlap jadwal workstation** (workstation tidak bisa dipakai dua job melebihi kapasitas).

7.3 Stock Entry untuk Produksi

Purpose	Kapan dipakai
Material Transfer for Manufacture	Memindahkan bahan dari gudang ke WIP
Material Consumption for Manufacture	Konsumsi bahan bertahap (jika Continuous Consumption aktif)
Manufacture	Mencatat barang jadi & konsumsi bahan
Disassemble	Membongkar barang jadi
Material Transfer (return)	Mengembalikan sisa bahan

7.4 Downtime Entry

Menu **Tools → Downtime Entry**: catat Workstation, From/To Time, Operator, **Stop Reason** (Excessive Machine set up time, Unplanned machine maintenance, On-machine press checks, Machine operator errors, Machine malfunction, Electricity down, Other), Remarks. Data dipakai laporan **Downtime Analysis**.

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8. Perencanaan Material

8.1 Item Lead Time

Data waktu per item untuk MPS/MRP: Manufacturing Time (menit/unit), No. of Shift, Shift Time, No. of Workstations, Daily Yield, Capacity per Day, Purchase Time, Buffer Time.

8.2 Sales Forecast

1. **Sales Forecast → + Add**: Company, Posting Date, Parent Warehouse, From Date, Frequency, Demand Number.
2. Pilih item → **Generate Demand** (exponential smoothing dari histori Sales Order) atau isi manual.
3. Submit → status *Planned*; tombol **Create MPS** → status *MPS Generated*.

8.3 Master Production Schedule (MPS)

1. Isi Company, Parent Warehouse, From/To Date.
2. Ambil demand dari **Sales Forecast**, **Sales Orders** (Get Sales Orders), **Material Requests**.
3. **Get Actual Demand** → sistem menghitung kebutuhan per item per tanggal dengan lead time kumulatif.
4. Submit → MRP dihitung (background); lihat **Material Requirements Planning Report**.

8.4 Production Plan

1. **Production Plan → + Add**: *Get Items From* = Sales Order atau Material Request.
2. Filter (tanggal, customer, project, item) → **Get Sales Orders** / **Get Material Requests** → **Get Finished Goods for Manufacture**.
3. **Get Sub Assembly Items** (opsi: combine, skip yang stoknya tersedia, gudang sub-assembly).
4. **Get Raw Materials for Purchase** (opsi: include safety stock, consider minimum order qty, ignore existing ordered qty, include non-stock / subcontracted items, warehouse tertentu).

5. **Submit** → - **Make Material Request** → bahan yang kurang (Purchase/Transfer) - **Make Work Order** → WO untuk FG & sub-assembly (atau Subcontracting PO) - **Reserve Stock** (opsional)
6. Pantau status: *Not Started* → *Material Requested* → *In Process* → *Completed / Closed*. Laporan: **Production Plan Summary**.

Tombol **Download Materials Required** → file Excel kebutuhan bahan.

8.5 Blanket Order

Kontrak jangka panjang (dengan customer/supplier) berisi item, qty & rate yang disepakati dalam periode tertentu; Sales/Purchase Order dibuat dari Blanket Order dan qty-nya dipantau.

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9. Subcontracting (Maklon)

Alur:

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Subcontracting BOM ➞ Purchase Order (service) ➞ Subcontracting Order ➞ Kirim bahan (Stock Ent
                                                                        ➞ Subcontracting Receipt (
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Laporan: **Subcontract Order Summary**, **Subcontracted Raw Materials To Be Transferred**, **Subcontracted Item To Be Received**. Subcontracting per operasi juga bisa dibuat dari **Job Card**.

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10. Laporan & Dashboard

10.1 Laporan di Menu

Laporan	Tujuan	Filter utama
Production Planning Report	Cek kecukupan bahan untuk WO / SO terbuka	Company, Based On, Warehouse
Material Requirements Planning	Kebutuhan bahan hasil MPS	MPS, Company
Work Order Summary	Rekap WO: status, umur, qty	Company, tanggal, status
Production Analytics	Tren WO per periode	Company, range
Job Card Summary	Rekap Job Card & waktu	Company, tanggal, WO
Downtime Analysis	Downtime per alasan & workstation	Tanggal, workstation
Quality Inspection Summary	Rekap QC lulus/gagal	Tanggal
BOM Search	Cari BOM yang memakai item tertentu	Item
BOM Stock Report	Kecukupan stok untuk sebuah BOM	BOM, Warehouse, Qty (**lihat Issue I-01**)
BOM Operations Time	Waktu operasi per BOM	Item, BOM, Workstation
Work Order Consumed Materials	Bahan aktual terpakai per WO	Company, WO

10.2 Laporan Tersedia tetapi Tidak Ada di Menu

Akses lewat Search (⌘K / Ctrl+K) dengan nama laporan:

Laporan	Tujuan
BOM Explorer	Pohon BOM lengkap multi level
BOM Stock Analysis	Analisa stok terhadap BOM
BOM Variance Report	Selisih bahan aktual vs BOM
Process Loss Report	Susut proses per WO
Cost of Poor Quality Report	Biaya rework / corrective operation
Exponential Smoothing Forecasting	Peramalan permintaan
Completed / Open Work Orders, Work Orders in Progress	Daftar WO per status
Issued Items Against Work Order	Bahan yang dikeluarkan per WO
Work Order Stock Report	Stok vs kebutuhan WO
Production Plan Summary	Rekap progres Production Plan

10.3 Dashboard Manufacturing

Chart: Produced Quantity, Completed Operation, Work Order Analysis, Work Order Qty Analysis, Pending Work Order (umur), Job Card Analysis, Quality Inspection Analysis, Last Month Downtime Analysis. Number Card: Open WO, WIP WO, Manufactured Items Value, Monthly Total/Completed WO, Monthly Quality Inspection, Ongoing Job Card.

11. Contoh Skenario End-to-End

Produk: *Meja Kayu Jati 120cm* (FG). **Sub-assembly:** *Rangka Meja*. **Bahan:** Papan Jati, Kaki Meja, Sekrup, Cat Melamine.

Langkah	Dokumen	Isi
1	Item	Buat 4 bahan baku, 1 sub-assembly, 1 FG
2	Workstation	*Mesin Potong* (Rp 50.000/jam), *Meja Rakit* (Rp 30.000/jam), *Booth Cat* (Rp 40.000/jam)
3	Operation	Potong, Rakit, Finishing Cat
4	Routing	*Routing Meja*: Potong 20' → Rakit 30' → Cat 25'
5	BOM Rangka	2 Papan Jati + 4 Kaki + 16 Sekrup, operasi Potong & Rakit
6	BOM Meja	1 Rangka + 1 Papan Jati (top) + 0,5 L Cat, operasi Cat; Secondary: serbuk kayu (scrap)
7	Sales Order	Customer pesan 10 meja
8	Production Plan	Dari SO → Get Sub Assembly → Get Raw Materials → Make MR (bahan kurang) + Make WO
9	Purchase	MR → PO → Purchase Receipt bahan
10	Work Order Rangka	Start (transfer) → Job Card Potong & Rakit → Finish
11	Work Order Meja	Start → Job Card Cat → Quality Inspection → Finish
12	Delivery	Delivery Note & Sales Invoice dari SO
13	Analisa	Work Order Summary, Job Card Summary, BOM Variance, Process Loss

12. Troubleshooting & Pesan Error Umum

Pesan	Penyebab	Solusi
Work-in-Progress Warehouse is required before Submit	WIP Warehouse kosong	Isi di WO atau set default di Manufacturing Settings
Target Warehouse is required before Submit	FG Warehouse kosong	Isi Target Warehouse
Quantity to Manufacture must be greater than 0	Qty 0	Isi qty
Work Order cannot be raised against a Item Template	Item adalah template varian	Pilih item varian
BOM {0} must be active / must be submitted	BOM draft/nonaktif	Submit & aktifkan BOM
BOM {0} does not belong to Item {1}	Salah pilih BOM	Pilih BOM milik item tsb
Cannot deactivate or cancel BOM as it is linked with other BOMs	BOM dipakai BOM induk	Pakai BOM Update Tool → Replace
Operations cannot be left blank	With Operations dicentang tanpa operasi	Isi operasi / uncheck
Operation Time must be greater than 0	Waktu operasi 0	Isi Time in mins
Cost allocation between finished goods and secondary items should equal 100%	Alokasi biaya tidak 100%	Sesuaikan %
Completed Qty cannot be greater than 'Qty to Manufacture'	Over produksi	Naikkan Overproduction % atau koreksi qty
Stopped Work Order cannot be cancelled, Unstop it first	WO status Stopped	Re-open lalu cancel
Closed Work Order can not be stopped or Re-opened	WO sudah Closed	Buat WO baru
You can't make any changes to Job Card since Work Order is closed	WO closed	—
Row {0}: From time must be less than to time	Time log salah	Perbaiki jam
Capacity Planning Error...	Workstation tanpa jam kerja / waktu 0	Isi Working Hours, atau Disable Capacity Planning
Stok tidak cukup saat Manufacture	Bahan belum ditransfer / stok kosong	Cek Production Planning Report, transfer bahan dulu

13. Status Saat Ini, Issue & Gap

13.1 Status Situs Live (6 Nov 2026)

Aspek	Status
Onboarding	0/7 langkah — belum ada master data produksi
Transaksi	Open WO = 0, WIP WO = 0, Manufactured Value = Rp 0
Mata uang	Rupiah (Rp) sudah aktif
Branding	Kode sudah NexERP (`app_title = "NexERP"`), tetapi header sidebar situs live masih menampilkan **"ERPNext"**

13.2 Issue (bug / konfigurasi salah)

ID	Severity	Issue	Bukti	Dampak
I-01	Tinggi	Link **BOM Stock Report** di menu Reports & kartu Reports kemungkinan rusak; di kode hanya ada report **BOM Stock Analysis**	Sidebar & workspace JSON menautkan "BOM Stock Report"; yang ada hanya folder `report/bom_stock_analysis`	User klik → "Not Found"/error (perlu konfirmasi di live)
I-02	Sedang	Header sidebar situs live masih "ERPNext" padahal kode sudah rebrand	Screenshot vs `hooks.py`	Inkonsistensi branding; kemungkinan belum `bench migrate` / `clear-cache` /build di server
I-03	Rendah	Duplikasi menu: *Production Planning Report* muncul di Material Planning dan Reports; *BOM* di sidebar & kartu Subcontracting	workspace_sidebar JSON	Navigasi membingungkan
I-04	Rendah	Chart *Last Month Downtime Analysis* & *Quality Inspection Analysis* menyimpan filter tanggal statis (2019/2020) di fixture	Fixture `dashboard_chart`	Chart bisa kosong bila tidak di-refresh oleh `dashboard_fixtures.py`
I-05	Rendah	Terminologi terjemahan Bahasa Indonesia belum diverifikasi untuk istilah manufaktur (mix EN/ID di UI)	Screenshot UI masih EN	Adopsi user lapangan lebih lambat

13.3 Gap (fitur / proses yang belum ada atau belum terkonfigurasi)

ID	Area	Gap	Rekomendasi
G-01	Master data	Belum ada Item, Warehouse produksi, Workstation, Operation, BOM	Jalankan setup Bab 5; template import (Data Import)

ID	Area	Gap	Rekomendasi
G-02	Settings	Manufacturing Settings belum di-review (default gudang, backflush, capacity planning)	Workshop keputusan kebijakan (lihat 5.1)
G-03	Navigasi	Laporan bernilai (BOM Explorer, Process Loss, Cost of Poor Quality, BOM Variance, Production Plan Summary) tidak ada di menu	Tambah ke sidebar Reports
G-04	Navigasi	**Visual Plant Floor** hanya bisa diakses dari form Plant Floor; Blanket Order & Quality Inspection tidak ada di menu Manufacturing	Tambah link / shortcut
G-05	Home	Workspace tidak punya **Shortcut** (shortcuts = kosong); hanya 1 chart & 3 number card	Tambah shortcut: New Work Order, Job Card terbuka, Production Plan, Plant Floor; tambah card *Ongoing Job Card* & *Monthly Completed WO*
G-06	Dokumentasi	Belum ada SOP/manual Bahasa Indonesia & materi training per role	Dokumen ini + video/slide per role
G-07	Lokal Indonesia	Belum ada format cetak (Print Format) Bahasa Indonesia untuk Work Order, Job Card, Surat Perintah Kerja, Bukti Pengeluaran Bahan	Buat Print Format ID
G-08	Shop floor	Belum ada mode operator sederhana (tablet/barcode) di luar Visual Plant Floor	Evaluasi Visual Plant Floor + barcode Job Card; atau custom page operator
G-09	Kualitas	Belum ada Quality Inspection Template & parameter per item	Definisikan template QC per produk/operasi
G-10	Biaya	Belum ada Workstation Operating Component yang dipetakan ke akun GL (listrik, tenaga kerja, overhead)	Mapping COA biaya produksi (Indonesia)
G-11	Akses	Role belum dipisah (Planner / Supervisor / Operator / Gudang); Operator butuh akses Job Card	Buat Role Profile & User Permission per plant/gudang
G-12	Notifikasi	Belum ada notifikasi/Assignment Rule (WO terlambat, material kurang, downtime)	Notification & Assignment Rule
G-13	Testing	Belum ada UAT script & data dummy untuk skenario manufaktur NexERP	Buat UAT per skenario Bab 11
G-14	Integrasi	Belum ada integrasi mesin/IoT untuk status workstation & downtime otomatis	Fase lanjutan (API Workstation status)
G-15	Kustomisasi	Tidak ada custom field/script khusus NexERP di modul Manufacturing (masih 100% standar)	Kumpulkan kebutuhan industri klien

14. Next Action Development (Roadmap)

Fase 0 — Quick Fix (1 minggu)

#	Aksi	Terkait	Output
0.1	Verifikasi link BOM Stock Report di live; perbaiki link sidebar & workspace ke **BOM Stock Analysis**	I-01	Patch JSON workspace & sidebar

#	Aksi	Terkait	Output
0.2	Deploy rebrand: `bench build`, `bench migrate`, `bench clear-cache` di server	I-02	Header "NexERP"
0.3	Rapikan duplikasi menu	I-03	Sidebar bersih
0.4	Tambah laporan tersembunyi & Visual Plant Floor ke menu	G-03, G-04	Sidebar Reports lengkap
0.5	Tambah shortcut & number card di Home	G-05	Home lebih informatif

Fase 1 — Setup & Konfigurasi (2–3 minggu)

#	Aksi	Terkait
1.1	Workshop kebijakan Manufacturing Settings + dokumentasi keputusan	G-02
1.2	Template Data Import: Item, BOM, Workstation, Operation, Routing	G-01
1.3	Struktur gudang produksi per Company/plant	G-01
1.4	Mapping biaya Workstation → COA	G-10
1.5	Role Profile & User Permission	G-11
1.6	Quality Inspection Template	G-09
1.7	Selesaikan 7 langkah onboarding dengan data pilot	—

Fase 2 — Lokalisasi & Adopsi (2–4 minggu)

#	Aksi	Terkait
2.1	Review & lengkapi terjemahan Bahasa Indonesia istilah manufaktur (`erpnext/locale/id.po`)	I-05
2.2	Print Format ID: SPK/Work Order, Job Card, Bon Pengeluaran Bahan, Laporan Hasil Produksi	G-07
2.3	Notifikasi & Assignment Rule	G-12
2.4	Training per role + video singkat; distribusi manual ini	G-06
2.5	UAT dengan skenario Bab 11, sign-off	G-13

Fase 3 — Peningkatan (backlog)

#	Aksi	Terkait
3.1	Halaman operator (tablet, scan barcode Job Card, start/stop)	G-08
3.2	Dashboard KPI: OEE, yield, on-time WO, biaya per unit	—
3.3	Integrasi mesin/IoT untuk status & downtime otomatis	G-14
3.4	Fix chart fixture tanggal statis	I-04
3.5	Kustomisasi per industri klien (food: batch & expiry; garment: varian ukuran/warna; furniture: cutting list)	G-15
3.6	Automated test untuk kustomisasi NexERP	G-13

14.1 Prioritas Ringkas

Prioritas	Item
P0	I-01, I-02, G-01, G-02
P1	G-03, G-04, G-05, G-10, G-11, G-09
P2	I-03, I-05, G-06, G-07, G-12, G-13
P3	I-04, G-08, G-14, G-15

15. Lampiran

15.1 Daftar DocType Modul Manufacturing

DocType	Jenis	Keterangan
BOM	Submittable	Resep produk
BOM Creator	Submittable	Pembuat BOM bertingkat
BOM Update Tool	Single	Replace BOM / update biaya
BOM Update Log	Submittable	Log job BOM update
Blanket Order	Submittable	Kontrak jangka panjang
Downtime Entry	Normal	Catatan downtime
Job Card	Submittable	Kartu kerja operasi
Manufacturing Settings	Single	Pengaturan modul
Master Production Schedule	Normal	MPS
Operation	Normal	Master operasi
Plant Floor	Normal	Lantai produksi + visual
Production Plan	Submittable	Rencana produksi
Routing	Normal	Urutan operasi
Sales Forecast	Submittable	Peramalan permintaan
Work Order	Submittable	Perintah produksi
Workstation	Normal	Mesin / lini
Workstation Type	Normal	Kelompok mesin
Workstation Operating Component	Normal	Komponen biaya workstation
Item Lead Time *(modul Stock)*	Normal	Data lead time untuk MPS/MRP

Child table: BOM Item, BOM Operation, BOM Explosion Item, BOM Secondary Item, BOM Creator Item, BOM Update Batch, BOM Website Item/Operation, Job Card Item/Operation/Time Log/Scheduled Time/Secondary Item, Production Plan Item/Sales Order/Material Request/Sub Assembly Item/Item Reference/Material Request Warehouse, Material Request Plan Item, Master Production Schedule Item, Sales Forecast Item, Sub Operation, Work Order Item/Operation, Workstation Cost/Working Hour, Workstation Operating Component Account, Blanket Order Item.

15.2 Halaman (Page)

Page	Fungsi
bom-comparison-tool	Bandingkan dua BOM
visual-plant-floor	Monitor workstation (dipakai di form Plant Floor)

15.3 Lokasi File Konfigurasi (untuk developer)

Komponen	Path
Workspace Home	`erpnext/manufacturing/workspace/manufacturing/manufacturing.json`
Sidebar	`erpnext/workspace_sidebar/manufacturing.json`
Onboarding	`erpnext/manufacturing/module_onboarding/`, `onboarding_step/`
Number Card / Chart	`erpnext/manufacturing/number_card/`, `dashboard_chart/`, `dashboard_fixtures.py`
DocType	`erpnext/manufacturing/doctype/`
Report	`erpnext/manufacturing/report/`
Terjemahan ID	`erpnext/locale/id.po`

Dokumen ini disusun dari analisa kode sumber dan tampilan situs live. Item bertanda "perlu konfirmasi" harus diverifikasi langsung di server produksi.